

Problem

Develop the state equations for Prob. 16.12.

Step-by-step solution

Step 1 of 2

Refer to Figure 16.35 in the textbook.

The input excitation for the circuit is $8u(t)$.

The current through all the elements in a series circuit is same.

Write the expression for the voltage across the capacitor.

$$\begin{aligned} v_C(t) &= \frac{1}{C} \int i_C(t) dt \\ &= \frac{1}{1} \int i_C(t) dt \\ &= \int i_C(t) dt \\ &= \int i_L(t) dt \quad (\text{since, } i_C = i_L) \end{aligned}$$

Apply Kirchhoff's voltage law to the circuit.

$$\begin{aligned} 8u(t) - (1)i_R(t) - \left(\frac{1}{1}\right) \int i_C(t) dt - (1) \frac{di_L(t)}{dt} &= 0 \\ 8u(t) - i_L(t) - \int i_L(t) dt - \frac{di_L(t)}{dt} &= 0 \end{aligned}$$

Substitute the expression of $v_C(t)$ in the KVL equation.

$$8u(t) - i_L(t) - v_C(t) - i'_L(t) = 0$$

Here,

$$\frac{di_L(t)}{dt} = i'_L(t)$$

Simplify the expression.

$$i'_L(t) = 8u(t) - i_L(t) - v_C(t)$$

Step 2 of 2

Find the differentiation of the voltage across the capacitor.

$$\frac{dv_C(t)}{dt} = \frac{d}{dt} \left[\int i_L(t) dt \right] \quad (\text{since, } i_C = i_L)$$
$$v'_C(t) = i_L(t)$$

Write the equations of voltage and currents in the state model.

$$\begin{bmatrix} v'_C(t) \\ i'_L(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 8 \end{bmatrix} u(t)$$

The expression for the output is,

$$v_o(t) = v_C(t)$$

Write the equation of voltage and currents in the state model.

$$[v_o(t)] = [1 \quad 0] \begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix}$$

Therefore, the required state equations for the circuit are,

$$\boxed{\begin{bmatrix} v'_C(t) \\ i'_L(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 8 \end{bmatrix} u(t)}$$
$$\begin{bmatrix} v_o(t) \end{bmatrix} = [1 \quad 0] \begin{bmatrix} v_C(t) \\ i_L(t) \end{bmatrix}$$